Ex.No:10	Web Application Security Testing : Exploiting SQL Injection and Reflected XSS Vulnerabilities
Date:	

Aim:

To understand and perform **web application vulnerability testing using Burp Suite**, and to identify and exploit common vulnerabilities such as **SQL Injection and Reflected Cross-Site Scripting (XSS)** using PortSwigger Web Security Academy labs.

Steps:

Procedure

Step 1: Install Burp Suite

1. Download **Burp Suite Community Edition** from the official PortSwigger website.
2. Run the installer and complete the installation process.
3. Launch **Burp Suite** on the system.

Step 2: Start a Temporary Project

1. When Burp Suite opens, select **Temporary Project**.
2. Click **Next**.
3. Choose **Use Burp Defaults**.
4. Click **Start Burp** to launch the interface.

Step 3: Open Burp Browser

1. Go to the **Proxy** tab in Burp Suite.
2. Click **Open Browser** to launch the built-in Chromium browser.
3. Ensure **Intercept is ON** so Burp can capture HTTP requests.

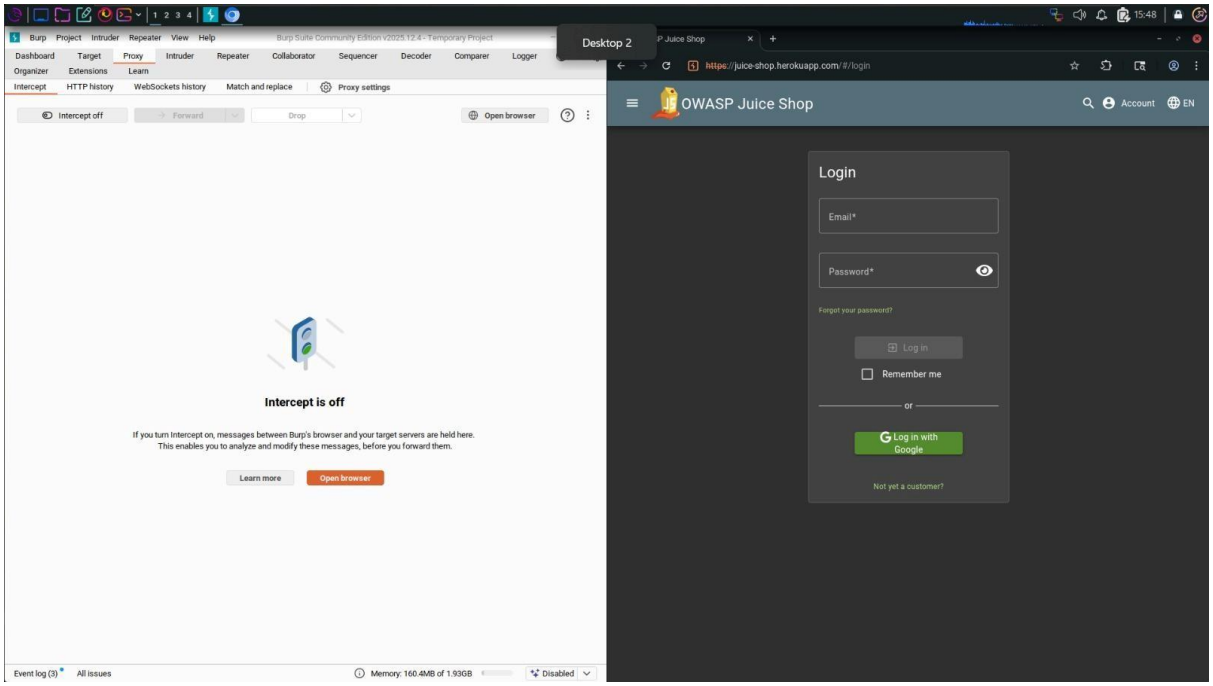
Step 4: Access the Vulnerable Web Application

1. In the Burp browser, open the vulnerable application **OWASP Juice Shop**.
2. Navigate to the **Login page**.

Step 5: Capture Login Request

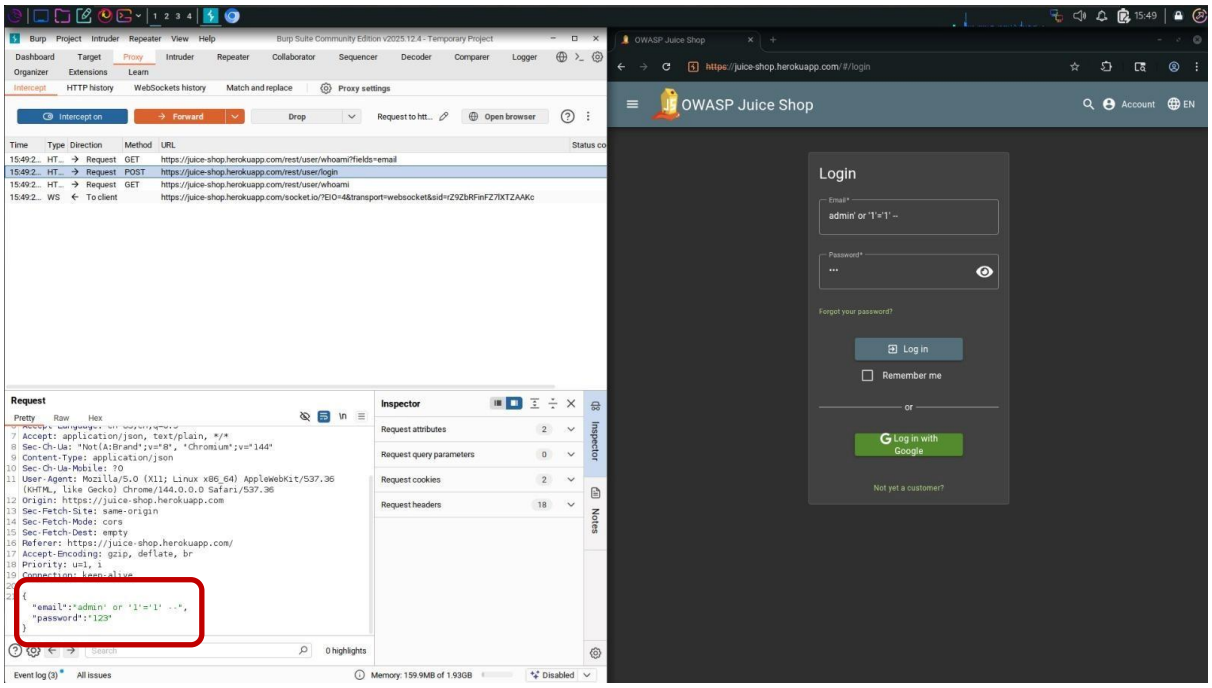
1. Enter any login credentials in the **Email** and **Password** fields.

2. Click **Log In**.
3. Burp Suite will intercept the HTTP request in the **Proxy → Intercept** tab.



Step 6: Perform SQL Injection

1. In the intercepted request, locate the **email parameter**.

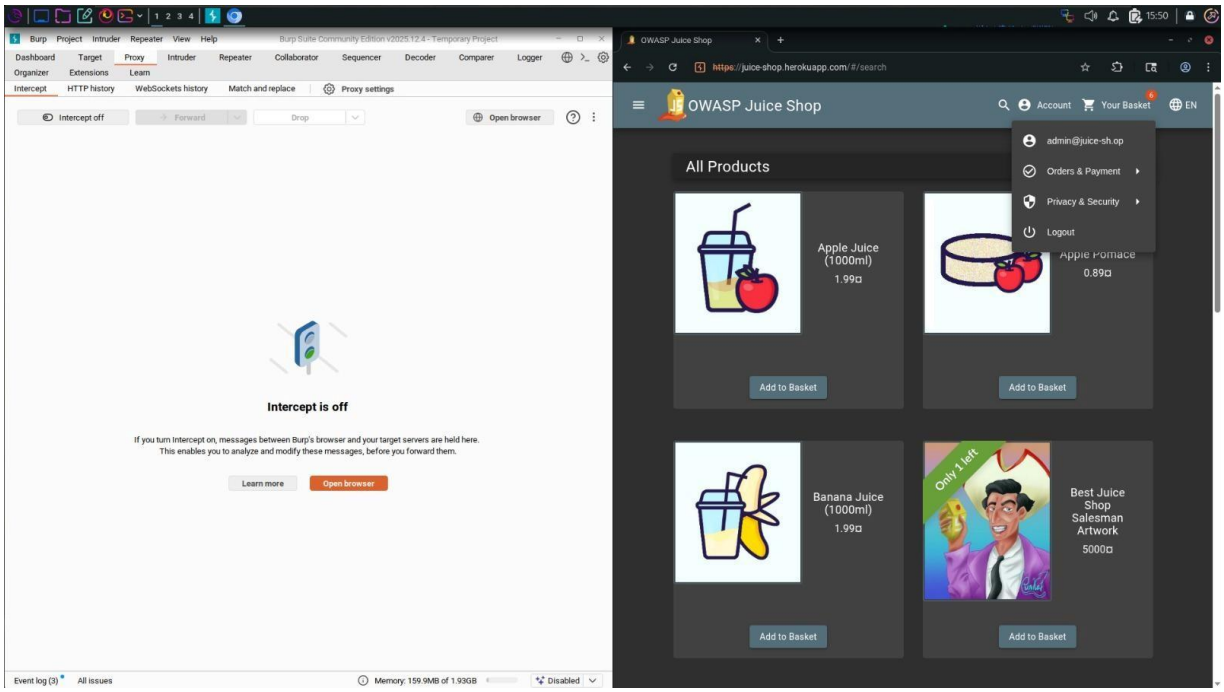


2. Modify the input with the SQL Injection payload:

admin' OR '1'='1' --

3. Forward the request using **Forward** in Burp Suite.

4. Observe that authentication is bypassed and access is granted.

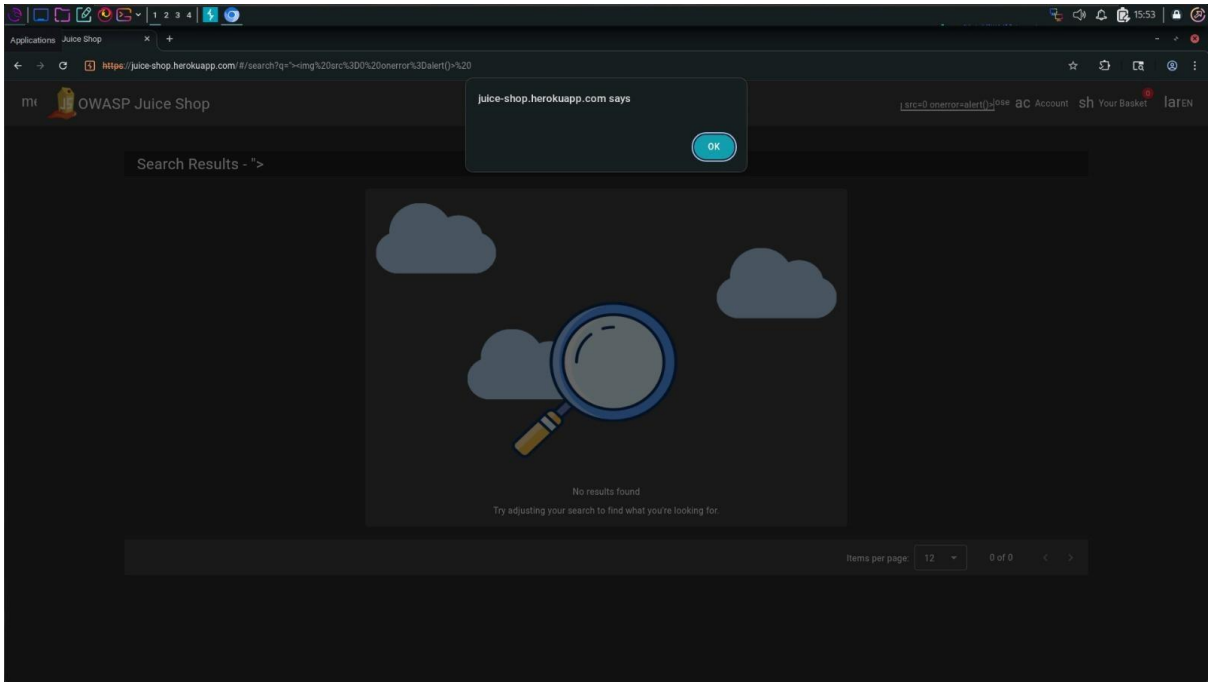


Step 7: Test for Cross-Site Scripting (XSS)

1. Navigate to the **Search bar** in the application.
2. Enter the following XSS payload:

   <img src=x onerror=alert()>

3. Submit the search request.



4. If a **JavaScript alert box** appears, it confirms a **Reflected XSS vulnerability**.

Step 8: Analyze Results

1. Observe the responses captured in Burp Suite.
2. Confirm the successful exploitation of **SQL Injection** and **XSS vulnerabilities**.

Result:

The web application was successfully tested using Burp Suite, and vulnerabilities such as **SQL Injection** and **Reflected Cross-Site Scripting (XSS)** were identified and exploited.